PCCoE	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering (PCCoE) (An Autonomous Institute) Affiliated to Savitribai Phule Pune University (SPPU)	

Group No- 13

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Lab 1: Literature Review

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Topic: BhashaBridge: An Inclusive Bi-Directional Accessibility Platform for Marathi Sign Language Interpretation and Automated Assistance.

1. Introduction

Communication is a basic human right, but the speech and hearing-impaired community often endures systemic isolation, kind of because there is this linguistic gap between signers and non-signers. Over the last ten years, progress in artificial intelligence has really sped up the making of automated sign language translation systems. Still, most of these solutions are basically designed for Western audiences only, like American Sign Language (ASL), or they depend on costly and kind of restrictive setups such as depth-sensing cameras or wearable kinetic gloves.

Also, the current systems usually work like stubborn, passive dictionaries, where one gesture equals one word, and they do not truly consider grammar in motion, or the real conversational intent of the person using it. What's missing, and it's still not addressed, is a clear gap in regional Indian linguistics, especially for Marathi Sign Language (MSL). This project suggests a real change in approach, by building a hardware-independent, bi-directional communication platform. Using YOLO-based spatial object detection, Large Language Models (LLMs) for contextual natural language processing, and agentic AI workflows, the system is intended to move sign language technology from a passive translation tool into a more active, autonomous digital assistant, meant for the MSL community.

2. Search Strategy

To get a solid theoretical base for this work, we did a fairly detailed literature search. The focus was on recent progress in computer vision, neural machine translation, and assistive tech in general, even if some terms overlap a bit. Overall it was kind of broad, but still targeted.

Keywords: sign language recognition ; real-time bidirectional translation ; YOLOv8 gesture detection ; Marathi Sign Language ; cloud native microservices ; LLM natural language processing ; Agentic AI assistive technology ; low-resource languages.

Databases Used: IEEE Xplore, PubMed/PMC, Springer Link, ScienceDirect, Wiley Online Library, Nature (Scientific Reports), Frontiers, and also Google Scholar.

Publication Years: 2023 - 2026

Inclusion Criteria: we only took peer-reviewed journal or conference papers from 2023–2026. These had to use deep learning / AI for sign language translation, or bidirectional communication loops, plus natural language processing for low-resource languages. We also included papers that discuss microservice architectures. Only papers available in English were considered.

Exclusion Criteria: we removed anything published before 2023. Also excluded were studies that used purely manual or statistical translation, meaning non-AI approaches. Papers that depended too much on very specific outside hardware (for example flex-sensor gloves) without any camera based alternative were not used either. Duplicate items, or sources that were not peer-reviewed, were also left out.

No of Papers Found: around 140, across all the keyword queries and databases

No of Papers Selected: 15

3. Selected Research Papers

The Table below lists the 15 research papers selected for detailed review, covering spatial object detection, contextual natural language generation, and bidirectional communication frameworks directly relevant to BhashaBridge.

SR No.	Title	Authors	Year	Publisher
1	GestureSpeak: A Real-Time Sign to Speech Translation	Oum Tipare, Saniya Pathre, Parth Pathak, Deepak Karia	2025	IEEE (Conference Paper)
2	A Prototype for Mexican Sign Language Recognition and Synthesis in Support of a Primary Care Physician	Candy Obdulia Sosa-Jimenez, Homero Vladimir Rios-Figueroa, Ana Luisa Solís-González-Cosío	2022	IEEE Access
3	Indian Sign Language Alphanumeric Hand Gestures Recognition Using Deep Learning Techniques	Shakana Mahendran, Ramya Mahendran	2024	IEEE
4	Sign Language Recognition and Translation to Text and Audio for Divyang Communication as a Microservice	D. M. Potey, M. Yasmin, Yash Dhaytonde, et al.	2025	IEEE (International Conference on Electronics)
5	Sign Language Recognition Using Artificial Intelligence and Machine Learning	Various Authors (et al.)	2025	Springer - Journal of Imaging Informatics in Medicine
6	Word Level Ethiopian Sign Language Recognition Using Hybrid CNN-LSTM Model	Various Authors (et al.)	2025	Springer - BMC
7	A Framework for Video Based Sign Language Interpretation Using Machine Learning and Statistical Methods	Various Authors (et al.)	2024	Research Journal
8	Real-Time Bidirectional Assistive Communication System for Speech and Hearing-Impaired Persons Using Deep Learning Models	Vellanki Navya Bharathi, Shaik Rizwan, Rakesh Reddy Gangavarapu, Aditya Thangapandi	2025	IEEE (International Conference on Emerging Technologies)
9	Conversion of Indian Sign Language to Speech by Using Deep Neural Network	Various Authors (et al.)	2022	IEEE Access
10	Real-Time Bidirectional Sign Language Translation System Using Machine Learning	Various Authors (et al.)	2025	Research Journal
11	A Portable System for Bidirectional Sign Language Translation for Deaf and Dumb	Various Authors (et al.)	2023	IEEE (Conference Paper)
12	Use of Neural Machine Translation in Multimodal Translation	Various Authors (et al.)	2023	IEEE Access
13	Indian Sign Language Recognition Using Convolutional Vision Transformers	Ankan Dutta, Spandan Priyam Chetia, Mwkthangsa Daimari, K. Suganya Devi	2025	IEEE Access
14	Leveraging Natural Language Processing for Real-Time and Accurate Speech-to-Sign Language Translation	Pooja A. Bagane, Sonali Kothari, Aradhana Pratap, Amol V. Dhumane	2024	IEEE (International Conference on Computing)
15	Hybrid Deep Learning and Fuzzy Matching for Real-Time Bidirectional Arabic Sign Language Translation	Mogeeb A. A. Mosleh, Ahmed A. A. Mohammed, Ezzaldeen E. A. Esmail, Rehab A. A. Mohammed, Basheer Almuhaya	2025	IEEE Access

4. Paper-wise Summary

Paper 1: GestureSpeak: A Real-Time Sign to Speech Translation – Tipare et al., 2025

Problem: Communication between hearing or speech-impaired people and the general public is pretty difficult, mainly because there aren't solid real-time sign language translation systems.

Methodology: We developed a real time system that takes in hand gestures from a webcam, then the gesture gets recognized with deep learning and finally it converts into speech instantly, like in the same moment.

Dataset: We built a custom gesture dataset, the sign images were collected under different conditions so it can cope with variety a bit.

Algorithms: A CNN based gesture recognition model, plus a speech synthesis module, working together.

Results: The system reached accurate real-time gesture recognition, low latency, and it generated speech that was actually understandable.

Limitations: Performance drops when the lighting is poor, and also when the gestures are more complex or especially dynamic, it struggles more.

Key Learning: A CNN based real-time gesture recognition stack, paired with speech synthesis, really looks like an effective assistive communication approach.

Paper 2: A Prototype for Mexican Sign Language Recognition and Synthesis in Support of a Primary Care Physician – Sosa-Jimenez et al., 2022

Problem: Healthcare professionals often run into communication barriers with deaf patients, mostly because they have limited knowledge of sign language, so it is hard to get the message across.

Methodology: They designed a prototype, able to recognize Mexican Sign Language gestures, and then translate those into text, and also produce synthesized speech for clinical communication , basically for the doctor-patient interaction.

Dataset: The dataset was a Mexican Sign Language gesture collection, gathered specifically for healthcare scenarios.

Algorithms: Deep learning based image classification paired with speech synthesis, yes, that combo again.

Results: The prototype showed effective communication support between physicians and hearing impaired patients, in the scenarios they tested.

Key Learning: It had a limited vocabulary, and the evaluation was done only in controlled environments, so outside that it might be different.

Paper 3: Indian Sign Language Alphanumeric Hand Gestures Recognition Using Deep Learning Techniques – Mahendran et al., 2024

Problem: Automatic recognition of Indian Sign Language (ISL) alphanumeric gestures is still pretty hard because gestures look too similar.

Methodology: The work trained deep learning models to sort ISL alphabets and numbers, using hand gesture images, as the input.

Dataset: They used an Indian Sign Language alphanumeric gesture dataset, specifically for that purpose.

Algorithms: CNN and other deep learning image classification approaches were applied.

Results: The models reached strong recognition accuracy for letters, and also numerical gestures.

Limitations: It was mostly limited to static hand gestures ,and the background had to be controlled or clean.

Key Learning: Deep learning can really enhance recognition accuracy, for ISL alphabets and numbers overall.

Paper 4: Sign Language Recognition and Translation to Text and Audio for Divyang Communication as a Microservice – Potey et al., 2025

Problem: Many current sign language systems are awkward to plug into modern applications in a smooth way.

Methodology: They built a microservice based setup that detects sign language, then translates it into text and audio, kinda like a service pipeline.

Dataset: They relied on a custom sign language dataset they prepared.

Algorithms: CNN style gesture recognition was used, and the system was wrapped with REST microservices.

Results: They managed to convert sign gestures into text and speech, and also kept it scalable because microservices were used.

Limitations: One catch is that a stable network infrastructure is needed, especially for distributed deployment .

Key Learning: Using microservice architecture helps scalability, and makes integration easier for sign language systems.

Paper 5: Sign Language Recognition Using Artificial Intelligence and Machine Learning – Springer, 2025

Problem: Traditional communication methods just don't support seamless interaction for hearing-impaired users, the gap is kinda persistent.

Methodology: We reviewed AI and ML approaches for sign language recognition, and we compared how well each one works in practice, kind of head to head.

Dataset: We used multiple benchmark sign language datasets .

Algorithms: CNN, RNN, LSTM, SVM, and Random Forest, all mixed in the comparison.

Results: Deep learning models consistently beat conventional machine learning methods, by a noticeable margin, every time it shows up in the reported numbers.

Limitations: You really need large annotated datasets for top performance, there's no easy shortcut.

Key Learning: CNN and hybrid deep learning models give better recognition accuracy than the others, even if the setup is heavier.

Paper 6: Word Level Ethiopian Sign Language Recognition Using Hybrid CNN-LSTM Model – Springer, 2025

Problem: Recognizing full sign language words feels harder than just spotting single characters, like it's not as simple.

Methodology: The hybrid CNN- LSTM model was proposed to pull out spatial and temporal traits from sign videos, yeah.

Dataset: Ethiopian Sign Language word level video dataset.

Algorithms: A CNN with an LSTM hybrid setup, pretty standard but still useful.

Results: They reported better recognition accuracy for continuous word gestures, not perfect but improved.

Limitations: Still computationally expensive and it needs lots of large video datasets, so, a trade-off.

Key Learning: When CNN and LSTM are combined properly, the model captures both spatial and temporal signals.

Paper 7: A Framework for Video Based Sign Language Interpretation Using Machine Learning and Statistical Methods – 2024

Problem: Getting accurate interpretation from continuous sign language videos keeps being difficult.

Methodology: A video-processing framework was proposed, integrating machine learning with statistical assessment and feature handling, somewhat “hybrid” in practice.

Dataset: A video based sign language dataset.

Algorithms: Machine learning classifiers plus statistical feature extraction, basically they mix both angles.

Results: It showed good performance for interpreting continuous sign videos, compared to baseline approaches.

Limitations: However it is sensitive to background noise and also illumination changes, which can mess things up.

Key Learning: Using video-based feature extraction helps with continuous sign recognition, it gives the system more context.

Paper 8: Real-Time Bidirectional Assistive Communication System for Speech and Hearing-Impaired Persons Using Deep Learning Models – Bharathi et al., 2025

Problem: Most existing systems only let information go one way, so it feels kind of limiting, if you need real back-and-forth help.

Methodology: They built a bidirectional setup that turns speech into sign and then also sign back into speech, in real time, though the exact flow can feel a bit juggling at first, you know.

Dataset: They used custom datasets for both speech and sign language, collected specifically for this work, not just off-the-shelf.

Algorithms: CNN and NLP, plus speech recognition and text-to-speech, all tied together in some pipeline.

Results: This enabled two way communication , with high recognition accuracy overall, which is good in real use.

Limitations: When the sentences get more complex, the recognition accuracy drops a bit , so it struggles with nuance.

Key Learning: Overall it shows that bidirectional assistive communication is more practical, because it matches how people actually interact in daily life.

Paper 9: Conversion of Indian Sign Language to Speech by Using Deep Neural Network – IEEE Access, 2022

Problem: Relying on manual ISL interpretation reduces communication speed and efficiency, basically it slows things down for everyday conversation.

Methodology: They designed a deep neural network that reads ISL gestures and converts them into spoken output, so the listener gets speech instead of gestures.

Dataset: They trained on an Indian Sign Language gesture dataset.

Algorithms: Deep Neural Network, like a DNN, used for the gesture-to-speech conversion.

Results: The system gave accurate speech results based on the recognized gestures, so the output was meaningful.

Limitations: The approach mostly covered static gestures, so anything more dynamic or moving over time might not be handled as well.

Key Learning: It highlights how deep neural networks can boost speech generation from sign language, even if the starting points are visual gestures.

Paper 10: Real-Time Bidirectional Sign Language Translation System Using Machine Learning – 2025

Problem: Real-time communication requires both sign-to-text and text-to-sign translation.

Methodology: Implemented a real-time translation framework using machine learning techniques.

Dataset: Bidirectional sign language dataset.

Algorithms: CNN with machine learning classifiers.

Results: Successfully performed two-way translation with low latency.

Limitations: Vocabulary coverage remains limited.

Key Learning: Real-time bidirectional translation enhances accessibility for hearing-impaired users.

Paper 11: A Portable System for Bidirectional Sign Language Translation for Deaf and Dumb – 2023

Problem: Existing translation systems lack portability for everyday use.

Methodology: Designed a portable embedded device capable of bidirectional sign language translation.

Dataset: Custom portable-device gesture dataset.

Algorithms: CNN with embedded processing.

Results: Achieved real-time translation on portable hardware.

Limitations: Limited computational resources affect performance.

Key Learning: Portable systems increase the practicality of assistive technologies.

Paper 12: Use of Neural Machine Translation in Multimodal Translation – 2023

Problem: Conventional translation systems cannot efficiently process multimodal inputs.

Methodology: Applied Neural Machine Translation techniques to combine visual and textual information.

Dataset: Multimodal translation datasets.

Algorithms: Neural Machine Translation (Transformer-based).

Results: Improved multilingual translation quality.

Limitations: Requires extensive computational resources and large datasets.

Key Learning: Transformer-based NMT effectively supports multimodal translation tasks.

Paper 13: Indian Sign Language Recognition Using Convolutional Vision Transformers – Dutta et al., 2025

Problem: CNNs alone struggle with capturing long-range visual dependencies in sign language images.

Methodology: Proposed a hybrid Convolutional Vision Transformer model for ISL recognition.

Dataset: Indian Sign Language image dataset.

Algorithms: Convolutional Vision Transformer (CvT).

Results: Achieved higher accuracy than conventional CNN models.

Limitations: Requires powerful hardware and extensive training data.

Key Learning: Vision Transformers significantly improve sign language recognition performance.

Paper 14: Leveraging Natural Language Processing for Real-Time and Accurate Speech-to-Sign Language Translation – Bagane et al., 2024

Problem: Speech-to-sign translation requires accurate language understanding.

Methodology: Integrated NLP techniques with speech recognition to generate sign language output.

Dataset: Speech-language datasets with corresponding sign translations.

Algorithms: NLP, Speech Recognition, Deep Learning.

Results: Improved translation quality and contextual understanding.

Limitations: Performance depends on speech recognition accuracy.

Key Learning: NLP enhances semantic understanding in speech-to-sign translation.

Paper 15: Hybrid Deep Learning and Fuzzy Matching for Real-Time Bidirectional Arabic Sign Language Translation – Mosleh et al., 2025

Problem: Similar Arabic sign gestures often lead to misclassification.

Methodology: Combined hybrid deep learning with fuzzy matching to improve bidirectional sign language translation.

Dataset: Arabic Sign Language dataset.

Algorithms: CNN + LSTM + Fuzzy Matching.

Results: Improved recognition accuracy and reduced gesture ambiguity.

Limitations: Increased computational complexity compared to single-model approaches.

Key Learning: Hybrid deep learning with fuzzy matching provides robust real-time bidirectional translation by reducing gesture ambiguity.

5. Comparative Analysis

The reviewed literature can be grouped into four broad categories based on the application focus and deep learning techniques used for sign language recognition and translation, as summarized below.

Category	Typical Methods	Common Datasets	Reported Performance	Advantages / Limitations
Static Sign Language Recognition (Papers 1, 3, 9, 13)	CNN, Deep Neural Networks (DNN), Convolutional Vision Transformers (CvT)	Indian Sign Language (ISL), Marathi/Custom gesture image datasets	High recognition accuracy for static alphabets and gesture classification	Adv: Fast recognition, simple implementation, suitable for real-time applications. Lim: Limited to static gestures; performance affected by lighting and background variations.
Dynamic & Continuous Sign Language Recognition (Papers 6, 7, 15)	Hybrid CNN-LSTM, Video Processing, Fuzzy Matching	Ethiopian, Arabic and video-based sign language datasets	Improved recognition of continuous words and sequential gestures	Adv: Captures both spatial and temporal information, better for sentence-level recognition. Lim: Requires large video datasets and higher computational resources.
Bidirectional Communication & Translation Systems (Papers 4, 8, 10, 11, 14)	CNN, NLP, Speech Recognition, Text-to-Speech (TTS), Microservices	Custom sign language datasets with speech/text annotations	Real-time translation between sign language, text and speech with high usability	Adv: Enables two-way communication between hearing and speech-impaired users. Lim: Limited vocabulary, reduced accuracy for complex sentences, and dependency on speech recognition quality.

Across these three categories, Convolutional Neural Networks (CNNs) still feel like the most commonly used method for sign language recognition, mainly because they do a pretty solid job at pulling out the spatial patterns from hand gesture images, even when the motion is a bit messy. In newer papers, mostly 2024–2025, you can see CNNs mixed together with LSTM, Vision Transformers (ViT/CvT), Natural Language Processing (NLP), and also Neural Machine

Translation (NMT). The idea is to get better contextual awareness and more continuous gesture recognition, not just isolated signs.

Most real-time translation systems work in a straightforward way, they map sign language into text and speech successfully, and newer bidirectional setups can also support speech-to-sign translation, so the overall conversation feels more natural. This helps, especially for people with hearing impairments and speech related limitations, where the back and forth interaction matters a lot.

6. Research Gap

- Many current papers mostly treat static sign language recognition, but honestly there are fewer works that can handle dynamic gestures, continuous words, and full sentence level translation in real time, not just in a demo.
- Even if deep learning approaches like CNN , LSTM, Vision Transformers, and Neural Machine Translation have been used, only a small number of systems actually blend gesture identification, speech synthesis, text translation, and two way communication together as one end-to-end pipeline.
- Standardized public datasets are still not so easy to find, especially for Indian Sign Language ISL and other regional sign languages. A lot of researchers use their own custom data, so fair comparison and reproducibility get kind of messy.
- Most of the methods reviewed are tested inside clean laboratory setups, which means results often drop when moved to real environments—things like shifting lighting, complicated backgrounds, hand occlusion , and different camera viewpoints.
- Many real-time solutions also need heavy computation, so running them on low cost devices becomes difficult. Building lightweight models that stay very accurate for real-time usage is still an open research challenge.

7. Proposed Future Scope

- To create a sign language recognition and translation system in real-time that can recognize both static and dynamic gestures with improved accuracy.
- To implement sophisticated deep learning AI models like MediaPipe Hands, CNN, LSTM, and Vision Transformer to effectively extract both the spatial and temporal features.
- To design a bidirectional communication system that can facilitate Sign-to-Text, Sign-to-Speech, Speech-to-Text, and Speech-to-Sign conversion in a single app.
- To compile or create a large set of annotated Indian Sign Language covering alphabets, words, and also sentences under various conditions to make the model robust.
- To fine-tune it using lightweight deep-learning approaches so that it can be used in mobile devices or less powerful devices while preserving real-time functionality.
- To develop the system in such a way that it will be able to accommodate other local and foreign sign languages in order to improve the accessibility for communication groups.

8. Conclusion

The literature review shows that deep learning kinda pushed the whole area of sign language recognition and translation quite far, mainly because researchers have started using CNNs, LSTMs, Vision Transformers , Neural Machine Translation, and Natural Language Processing. A lot of current setups get strong accuracy for static hand gestures, but more recent work has drifted toward dynamic gesture recognition and even two way exchange between hearing-impaired and non-hearing-impaired people. That said , there are still a few issues that don't really go away. For example, standardized datasets are still scarce, recognizing continuous sign language is hard , many systems seem to only work well in controlled environments, and the computational load can be heavy for real-time use. Also, only a small number of systems actually bundle everything together, like gesture recognition, speech synthesis, text translation, and multilingual support, in one integrated pipeline. In this project, the main goal is to tackle those limitations by building an intelligent, real-time sign language recognition setup. The plan uses MediaPipe Hands for solid hand landmark extraction, deep learning models such as CNN and LSTM for recognizing the gesture sequence,

and speech synthesis so the communication sounds more natural. The system is meant to run efficiently in real-world conditions, while still being accurate, quick, and easy for users to interact with, especially for hearing- and speech-impaired individuals. Overall, by combining modern computer vision plus artificial intelligence approaches, this work is expected to help make communication more reachable, dependable, and inclusive.

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